

Rahul Shah

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EXPERIENCE

Software Engineer

May 2024 – Present

Solutions Unified LLC.

San Francisco, CA

- Built a documentation platform with an in-browser editor and automated sync that propagates Markdown diffs to **Amazon Knowledge Base**, powering a production **RAG chatbot** and reducing maintenance overhead by **60%**.
- Migrated manual customer environment provisioning (**EC2, S3, DynamoDB, Lambda**) to an automated **AWS CDK + CloudFormation** deployment framework integrated into the console, enabling self-serve environment setup and reducing onboarding time by **75%**.
- Designed a fault-tolerant, parameterized data ingestion and export pipeline using **Step Functions** and **Lambda** with scheduled workflows, idempotent execution, and automated failure handling reducing manual data operations by **70%**.
- Engineered an event-driven billing system where isolated environments publish usage metrics via EventBridge, **SNS**, and **SQS** to a central Lambda-Stripe integration eliminating manual reconciliation and reducing revenue leakage by **20%**.

AI Full-Stack Developer

Jan. 2024 – Jan. 2025

Enterprise Technology(Mastercard Foundation), Arizona State University.

Tempe, AZ

- Built core backend services for CreateAI, a secure **LLM platform serving 10k+ users**, enabling faculty to create custom AI chatbots with on-demand knowledge bases, role-based access control, and PII filtering guardrails.
- Integrated **Whisper (STT)** and **ElevenLabs (TTS)** into a voice-enabled LLM assistant achieving sub **500ms** bidirectional audio latency for real-time conversational interactions with LLMs.
- Designed a provider-agnostic **LLM abstraction layer** enabling dynamic model routing across 5+ vendors, supporting live **A/B experimentation** with zero downtime or code changes.
- Built a **multi-agent orchestration** framework using LangGraph and FastAPI with stateful memory, tool use, and retrieval automating complex multi-step workflows and cutting average task completion time by **45%**.

Software Engineer II (Full-Stack)

June 2020 – July 2023

Allegion (Schlage).

Bengaluru, India

- Designed and owned the lock gateway communication layer using **MQTT** and **WebSockets**, handling device state sync, command dispatch, and event streaming across **100k+** connected devices at production scale.
- Reduced average event processing latency by **81% (800ms to 150ms)** by architecting a **Kafka** and **Redis** based pipeline that decoupled telemetry ingestion for **50k+ concurrent devices**.
- Shipped a disposable one-time credential feature in the **mobile app** using time-bound **JWT tokens**, enabling users to generate temporary access keys for guests driving a **25%** increase in app adoption post-launch.
- Built and maintained a shared gateway **SDK** adopted across 3 internal teams, standardizing device authentication, command formatting, and error handling — reducing cross-team integration bugs by **40%**.

TECHNICAL SKILLS

Languages & Frameworks: Go, TypeScript/JavaScript, Python, FastAPI, React, Next.js, Node.js

AI & ML: Hugging Face, LangChain, AWS Bedrock, OpenAI, Whisper, Pinecone, RAG, Amazon Knowledge Base

Cloud & Infra: AWS (EC2, S3, ECS, Lambda, Step Functions, EventBridge, CDK), Kubernetes, Docker

Data & Messaging: PostgreSQL, DynamoDB, MongoDB, Redis, Kafka, SQS, SNS

PROJECTS

VoicePad ([link](#)) | Go, Python, TypeScript, PostgreSQL, React, Websockets, Ollama, gRPC

Nov. 2025 – Dec 2025

- Built a real-time collaborative whiteboard where spoken instructions are transcribed via Whisper, interpreted by Gemini Live API, and rendered as structured visual elements on a shared canvas — using gRPC for low-latency streaming between a Go backend and React frontend.

Conversense ([link](#)) ([Demo](#)) | Go, React, PostgreSQL, LiveKit, Google Gemini, AWS S3, Inngest

Oct 2025 – Nov 2025

- Built an AI coaching platform where users join live video calls with specialized AI agents (tutor, interviewer, sales coach) via LiveKit, featuring real-time sentiment analysis, automated transcription, post-call summarization, and a persistent chat interface to continue conversations with the agent after the call.

Vistruct ([link](#)) ([Demo](#)) | Python, React, TypeScript, FastAPI, PostgreSQL, Docker, GenAI

Jul. 2025 – Aug 2025

- Built a full-stack AI educational video platform adopted by 20+ professors, where user prompts are sent to an LLM to generate Manim animation scripts, executed in isolated containers with automated error re-prompting, and rendered videos stored to S3 with full versioning support.

EDUCATION

Arizona State University

Tempe, AZ

Master of Science in Computer Science GPA: 4.0/4.0

Aug. 2023 – May 2025